

LEVEL 1 (Q1-Q10)

Callback Basics

Q1

Create your own function:

```
greetUser(name, callback)
```

Output:

Hello Ram

Welcome Ram

Q2

Create:

```
calculate(a,b,callback)
```

Perform:

- Add
- Multiply
- Divide

Using callback.

Q3

Create a custom callback based calculator.

Input:

```
calculator(20,10,operation)
```

Output should depend on operation callback.

Q4

Create a function:

`processStudent(student, callback)`

Callback should print student details.

Q5

Create a function:

`downloadFile(callback)`

Simulate file download after 2 sec.

Q6

Create:

`registerUser(callback)`

After registration callback should send welcome email.

Q7

Create:

`placeOrder(callback)`

After order callback should generate invoice.

Q8

Create:

`fetchUser(callback)`

Return dummy user after 3 sec.

Q9

Create callback-based OTP verification.

Q10

Create callback-based login system.

LEVEL 2 (Q11-Q20)

setTimeout + Event Loop

Q11

Predict output

```
console.log("A");
```

```
setTimeout(()=>{  
  console.log("B");  
},0);
```

```
console.log("C");
```

Q12

Predict output

```
setTimeout(()=>{  
  console.log(1);  
},1000);
```

```
console.log(2);
```

Q13

Predict output

```
console.log("Start");
```

```
setTimeout(()=>{  
  console.log("Timeout");  
},0);
```

```
console.log("End");
```

Q14

Create countdown:

5

4

3

2

1

Boom

Using setTimeout.

Q15

Create digital bomb timer.

Q16

Create delayed greeting system.

User enters name.

After 2 sec greeting appears.

Q17

Create delayed notification system.

Q18

Create delayed button disable feature.

Q19

Create delayed redirect simulation.

Q20

Create a function:

`wait(seconds, callback)`

LEVEL 3 (Q21-Q30)

Callback Hell

Q21

Simulate:

Login

↓

Get User

↓

Get Orders

↓

Show Orders

Using nested callbacks.

Q22

Create callback hell with 5 nested levels.

Q23

Simulate food ordering system.

Order
↓
Cook
↓
Pack
↓
Deliver

Q24

Simulate hospital process.

Registration
↓
Doctor
↓
Test
↓
Medicine

Q25

Simulate college admission process.

Form
↓
Verification
↓
Payment
↓
Admission

Q26

Create callback chain for:

Signup

↓

Verify Email

↓

Login

↓

Dashboard

Q27

Create nested callbacks with random delays.

Q28

Create callback-based weather fetching simulation.

Q29

Create callback-based e-commerce checkout flow.

Q30

Create callback-based movie booking flow.

LEVEL 4 (Q31-Q35)

setInterval

Q31

Create live clock.

Q32

Create stopwatch.

Start

Pause

Reset

Q33

Create traffic light simulation.

Red

↓

Yellow

↓

Green

Repeat forever.

Q34

Create auto slider.

Every 3 sec image changes.

Q35

Create typing effect animation.

Example:

Hello World

One character at a time.

LEVEL 5 (Q36-Q40)

HARD EVENT LOOP QUESTIONS

Q36

Predict Output

```
console.log("A");
```

```
setTimeout(()=>{  
  console.log("B");  
},1000);
```

```
setTimeout(()=>{  
  console.log("C");  
},0);
```

```
console.log("D");
```

Q37

Predict Output

```
function one(){
```

```
  console.log("One");
```

```
}
```

```
setTimeout(one,0);
```

```
console.log("Two");
```

Q38

Predict Output

```
console.log("Start");
```

```
for(let i=0;i<10000000000;i++){}
```

```
setTimeout(()=>{  
  console.log("Timeout");  
},0);
```

```
console.log("End");
```

Q39

Create Event Loop Visualizer.

Show:

Call Stack

Web API

Callback Queue

Flow on screen using DOM.

Q40 🧠 FINAL BOSS

Build a Mini Async Task Manager